

NEXT • LIVE EXAMPLE

DEMO PPT

Watch the story become visible.

MODE

TRANSITION

FIELD

DARK STAGE

FOCUS

DEMO

REAL-TIME SIGN LANGUAGE TO TEXT & VOICE CONVERTER

Team Name: Iron Legions

Team Lead: Ayush Lahiri

Problem Statement: Detecting sign language hand gestures and converting them into real-time text and voice for inclusive communication

signetic.vercel.app

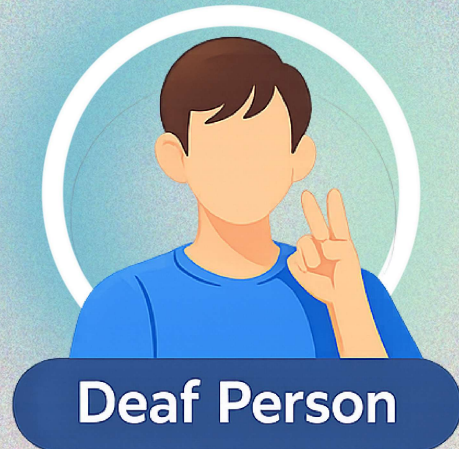


PROBLEM STATEMENT

Sign language users and non-signers lack a reliable, real-time communication method across digital platforms, leading to frequent misunderstandings and exclusion.

KEY ISSUES

- Specialized hardware dependency
- Cloud-based latency & privacy risks
- Poor real-world reliability
- Limited accessibility support
- Are not scalable, difficult to deploy for large scale use



INTRODUCING SIGNETIC

Signetic is a real-time sign language translation system that works directly in the browser to convert hand gestures into text and speech. It combines hand detection with pose-based gesture understanding to provide fast, stable, and privacy-preserving communication without special hardware.



InnovateX
Building the Future with Google Tech

KEY FEATURES OF SIGNETIC

ON-DEVICE EXECUTION

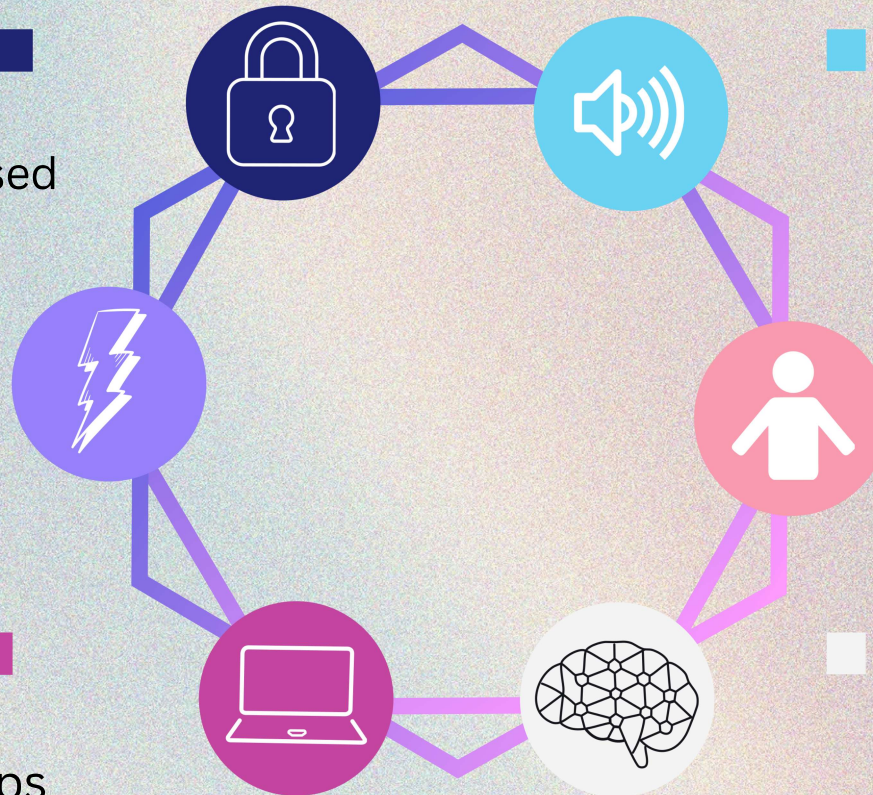
ensuring complete user privacy through cookie-based history-saving

REAL-TIME PERFORMANCE

optimized with WebGL acceleration

MOBILE RESPONSIVE UI

works smoothly across phones, tablets and desktops



SPEECH-ENABLED ACCESSIBILITY

for two-way communication

USER-FRIENDLY CONTROLS

clear, backspace, theme switcher and speech output

GESTURE CLASSIFICATION LOGIC

Each ASL alphabet is identified using rule-based geometric constraints

HOW DOES SIGNETIC WORK?

YOLOv8 Hand
Detection (Bounding
Box + Confidence)

Geometric Feature
Engineering

Temporal
Stabilization Engine

01

Webcam Feed

02

03

MediaPipe Hands
(21 × 3D Landmarks)

04

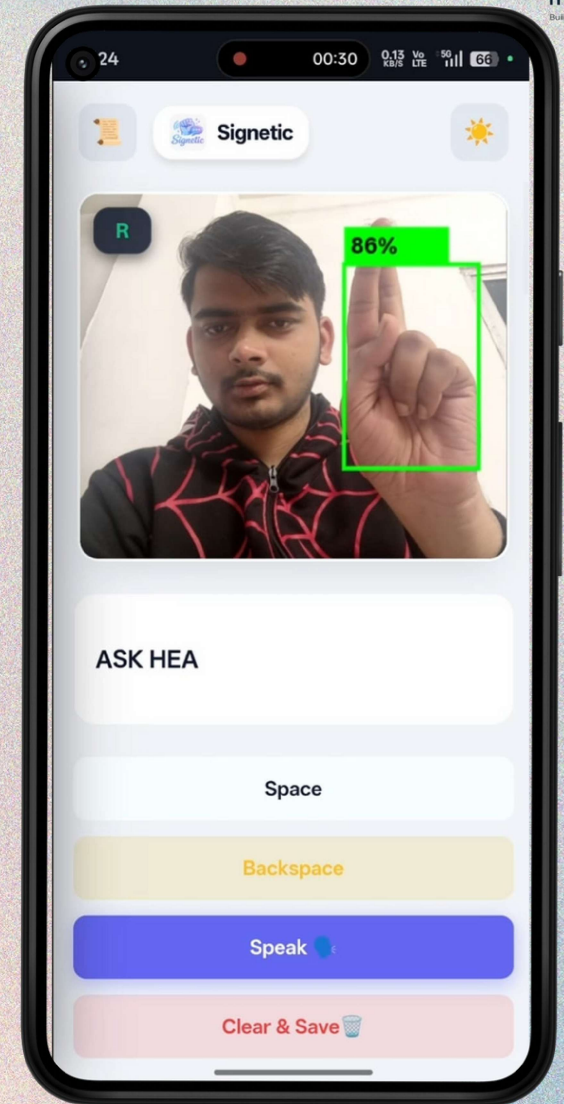
05

Rule-Based Gesture
Classification

06

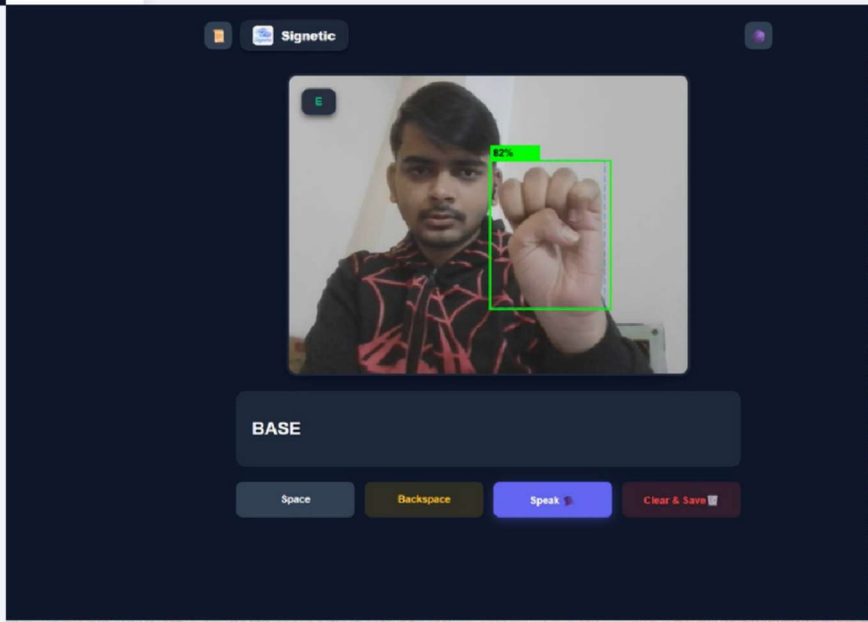
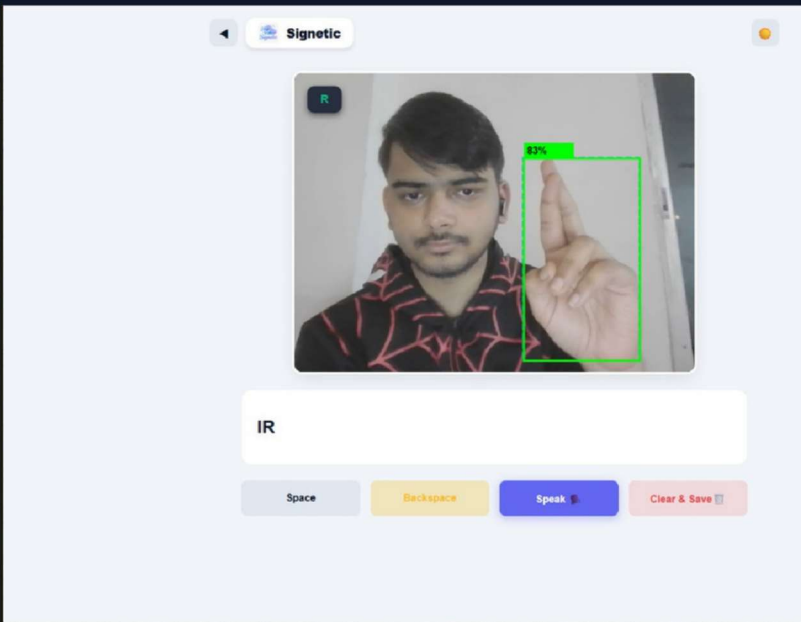
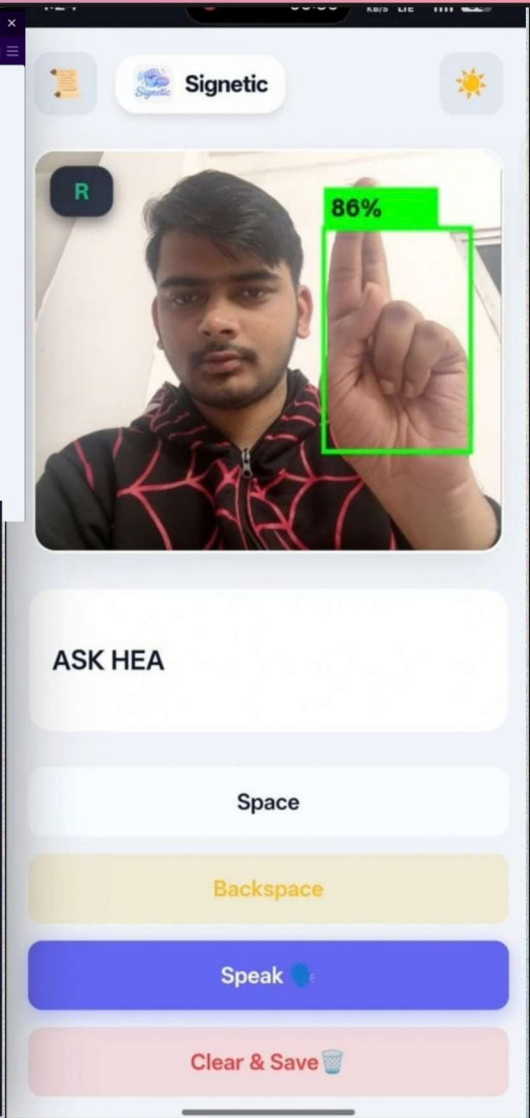
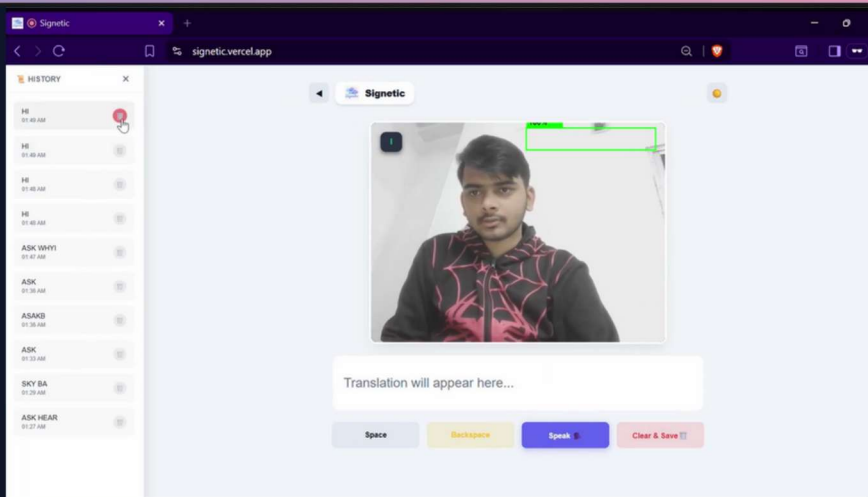
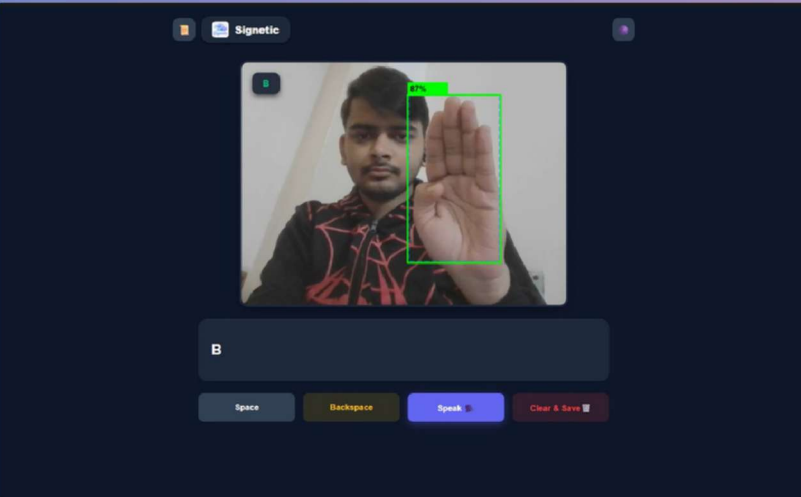
07

Text Output + Text-to-
Speech + history saving





SNAPSHOTS OF THE MVP





HI

Space

Backspace

Speak

Clear & Save

TEAM: IronLegions

2000000000

0:00



WHY IS SIGNETIC BETTER ?



HELLO

SIGNETIC

 **BROWSER-BASED**

 **NO SPECIAL HARDWARE**

 **PRIVACY-FIRST**

 **GEOMETRIC FEATURE ENGINEERING**

 **DUAL-MODEL (YOLOV8 + MEDIAPIPE)**

 **REAL-TIME, LOW LATENCY**

TRADITIONAL SIGN LANGUAGE CONVERTERS

 **CLOUD-BASED**

 **EXTRA DEVICES**

 **DATA SENT TO SERVERS**

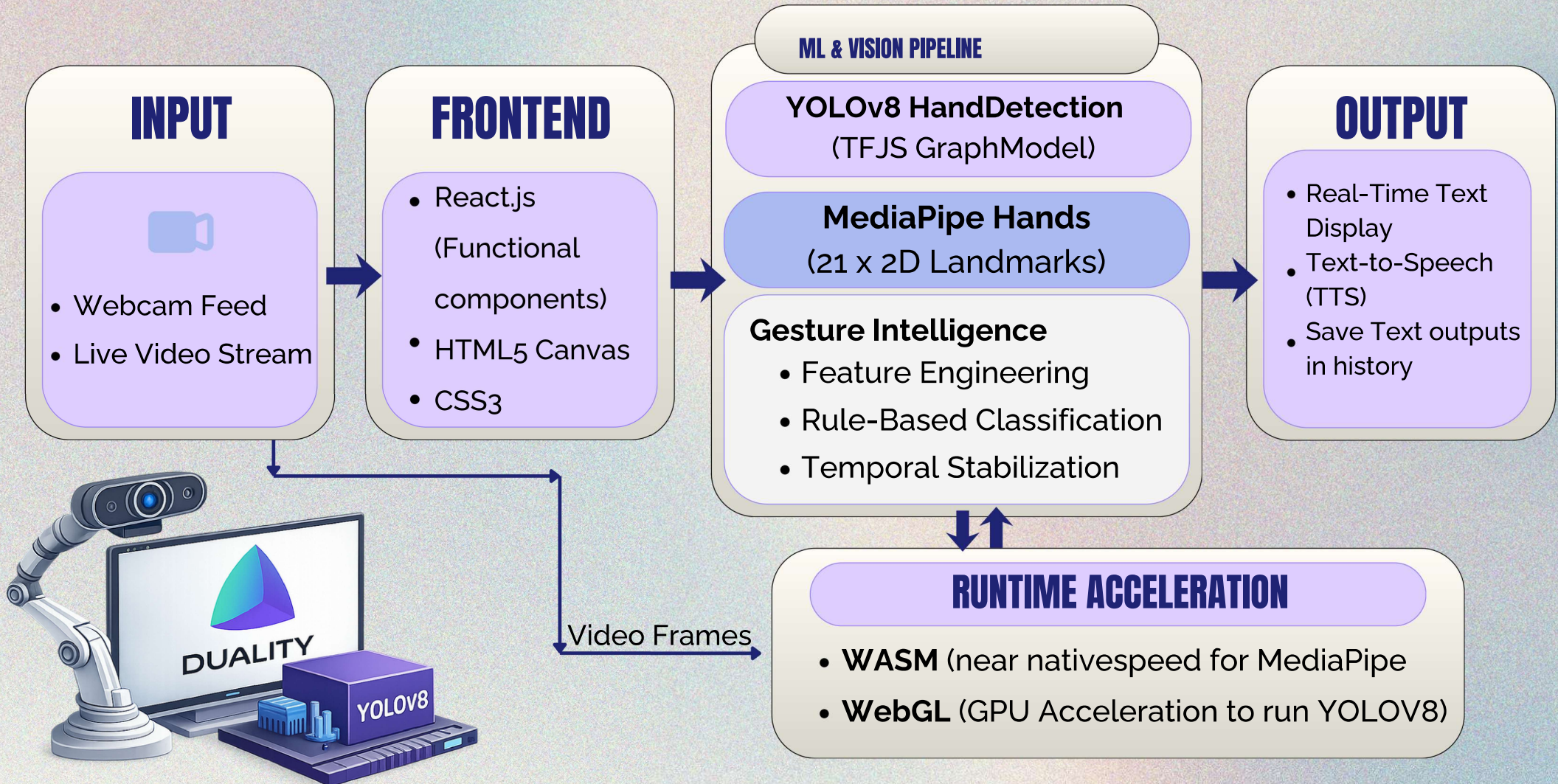
 **BLACK-BOX IMAGE CLASSIFICATION**

 **SINGLE-MODEL**

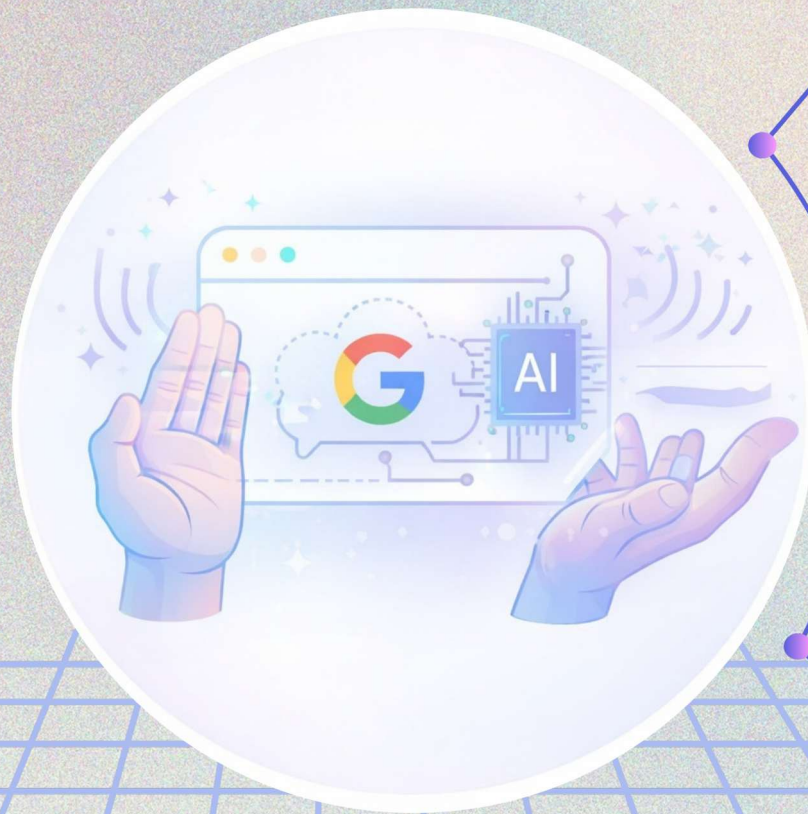
 **NOTICEABLE LAG**



ARCHITECTURE & PROCESS FLOW



GOOGLE TECH STACK USED



TensorFlow.js
Client-side ML inference



MediaPipe Hands
Real-time hand pose detection



WebGL
GPU acceleration

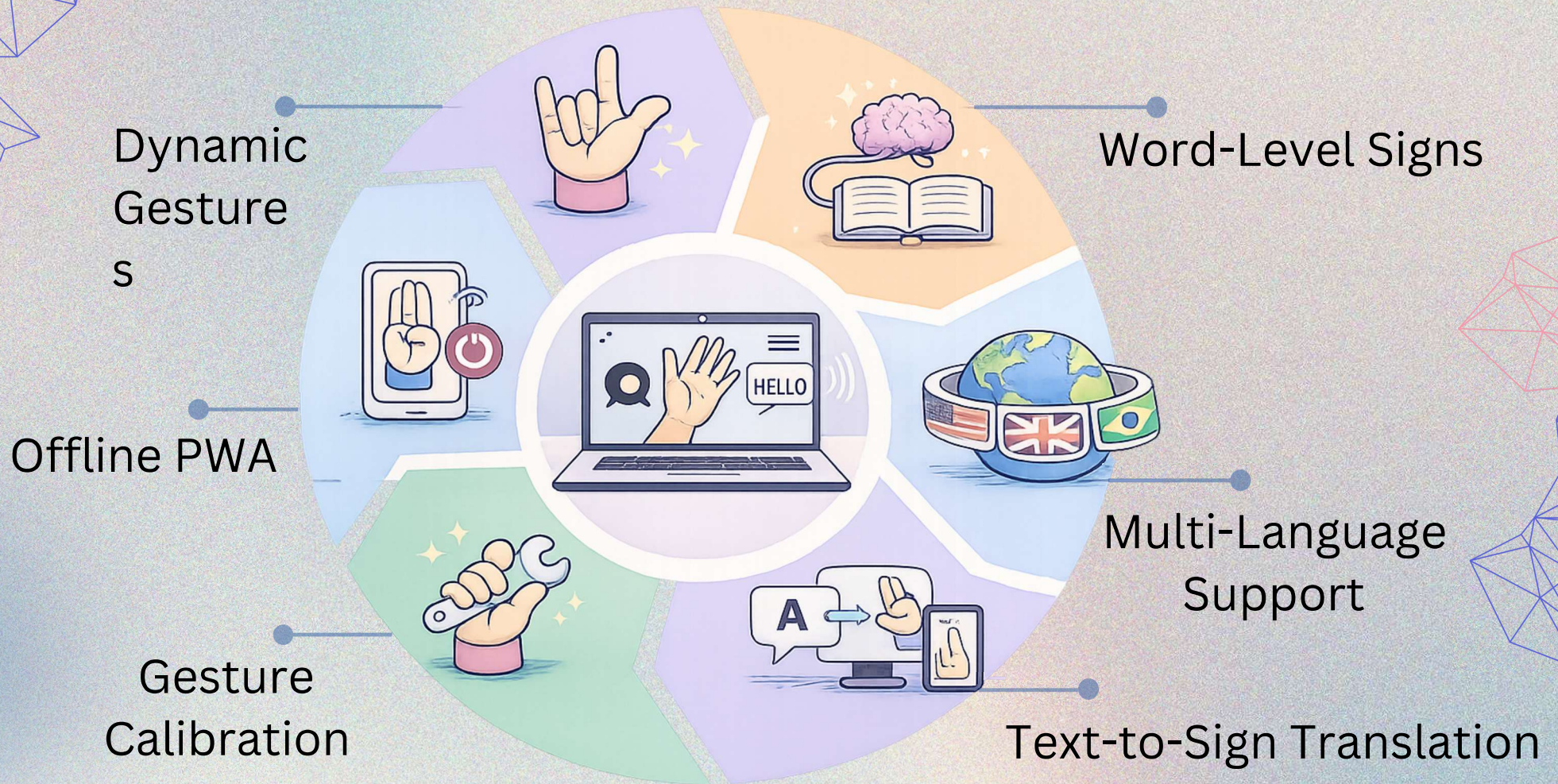


WebAssembly
High-performance execution



Web Speech API
Text-to-Speech

FUTURE SCOPE OF SIGNETIC





**THANK
YOU!**

PROJECT LINKS

- **GitHub Repository:**
<https://github.com/ayushL2007/IronLegions-Gesture>
- **Demo Video Link:**
<https://youtu.be/6M54ll2hhjk>
- **MVP Link:**
signetic.vercel.app